

SL-GPS Validation Report: GRI-Mech 3.0 Mechanism Reduction

SL-GPS Automated Validation

May 20, 2026

Abstract

Validation of the SL-GPS method on GRI-Mech 3.0 (53 species, 325 reactions) for CH_4/air autoignition. Training data generated over $T=900\text{--}2000\text{ K}$, $P=1\text{--}20\text{ atm}$. Validated on 6 cases spanning the parameter space.

1 Configuration

Table 1: Training parameters

Parameter	Min	Max
Temperature	900 K	2000 K
Pressure	1 atm	20 atm
X_{CH_4}	0.02	0.10
X_{O_2}	0.10	0.25
X_{N_2}	0.60	0.80
GPS α	0.001	
Training cases	20	

2 Validation Results

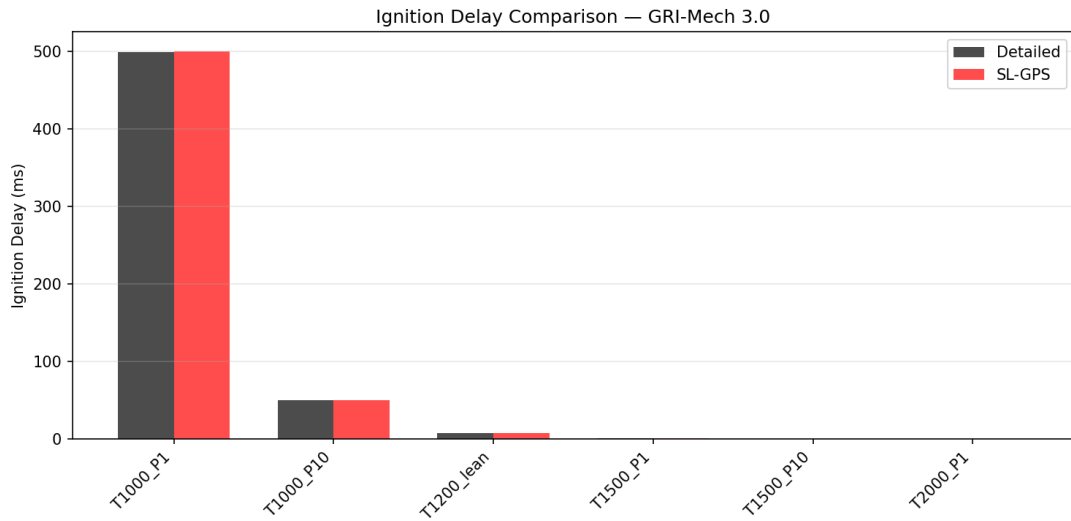


Figure 1: Ignition delay comparison: Detailed vs SL-GPS

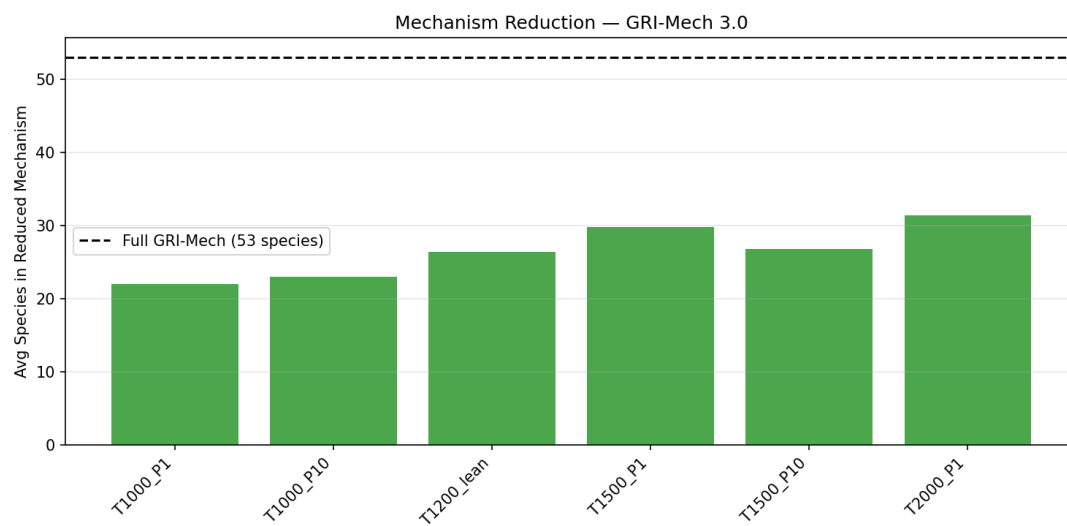


Figure 2: Average species count in reduced mechanism per case

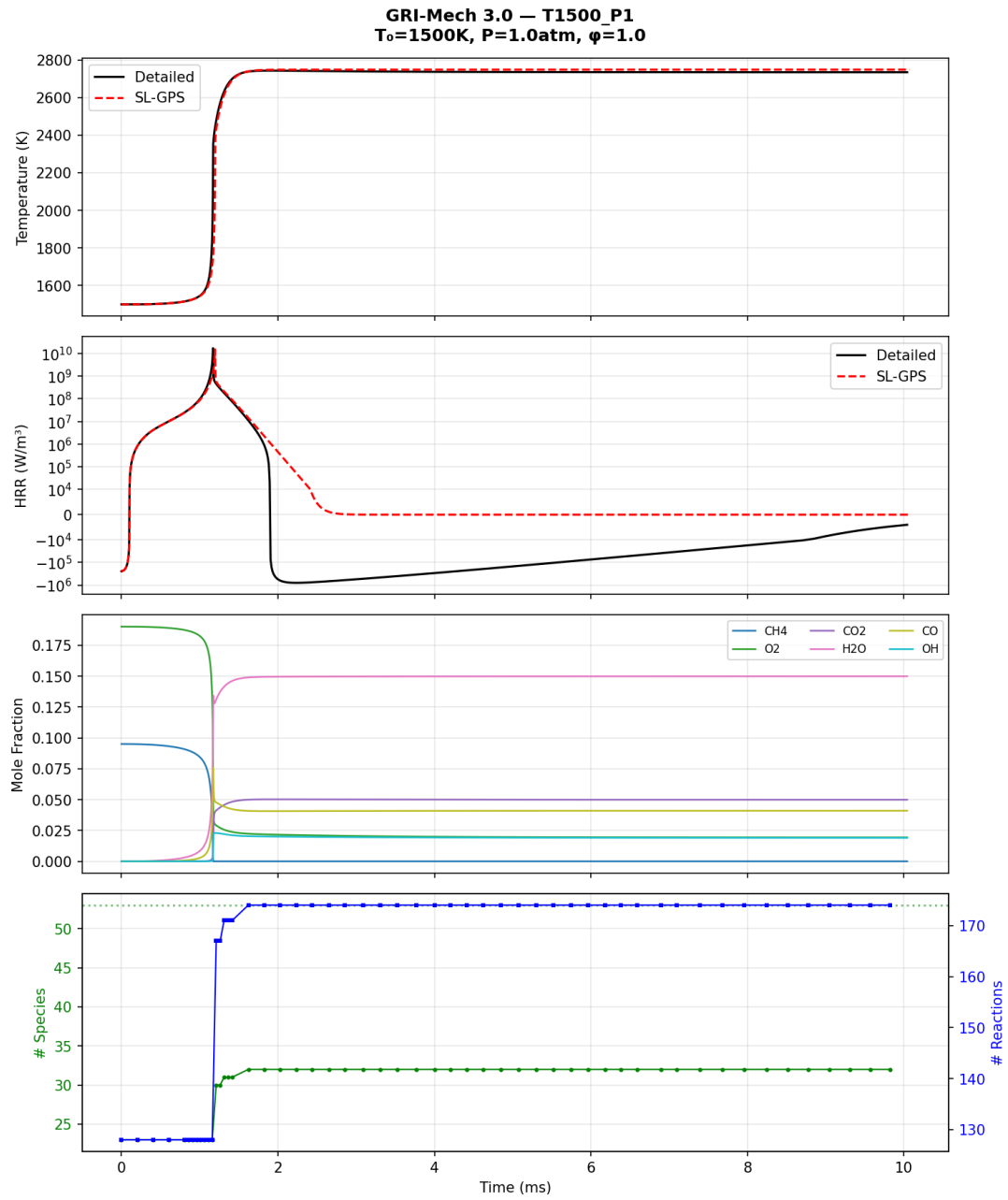


Figure 3: Case: $T=1500\text{K}$, $P=1\text{atm}$, $\phi=1.0$

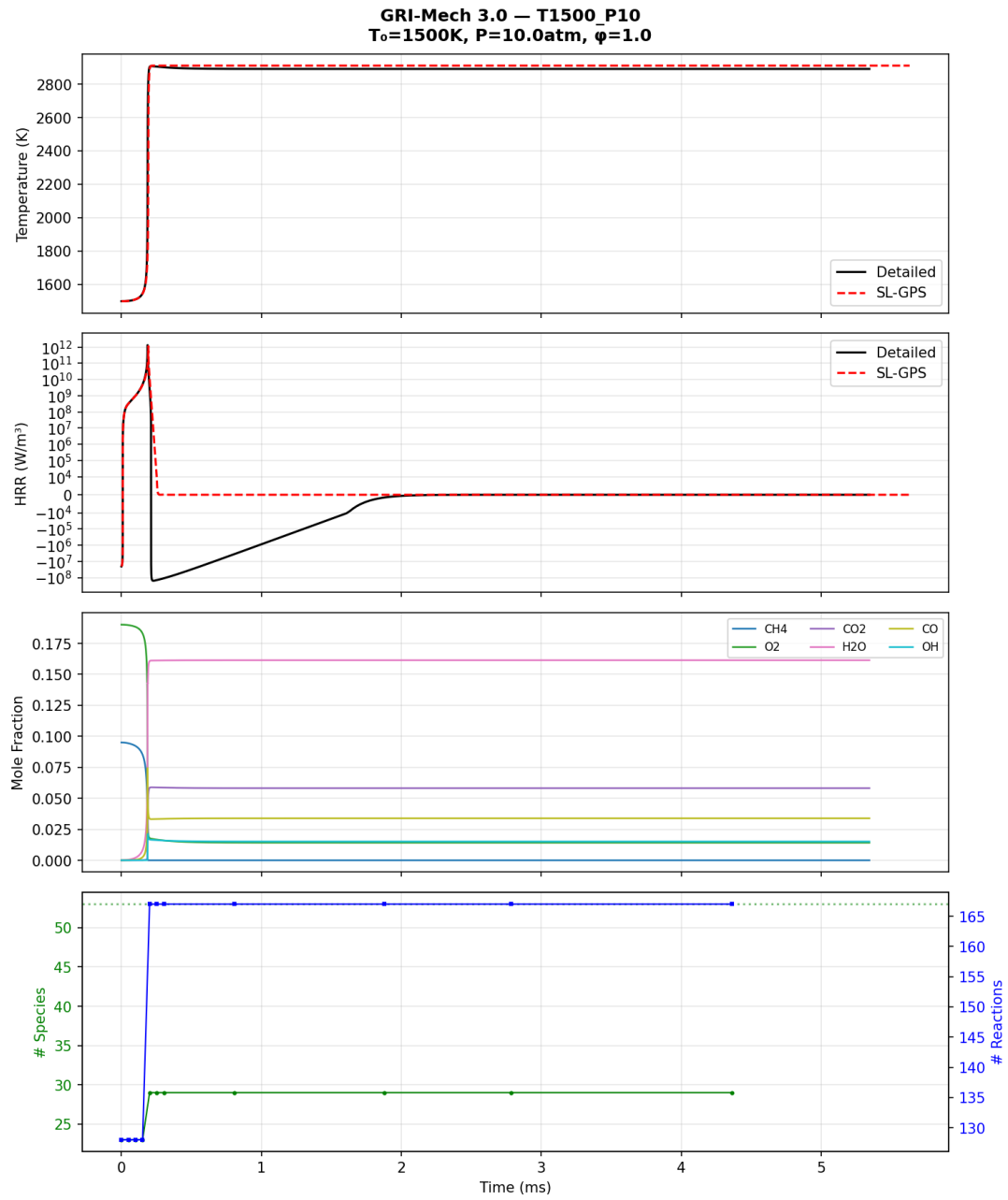


Figure 4: Case: T=1500K, P=10atm, ϕ =1.0

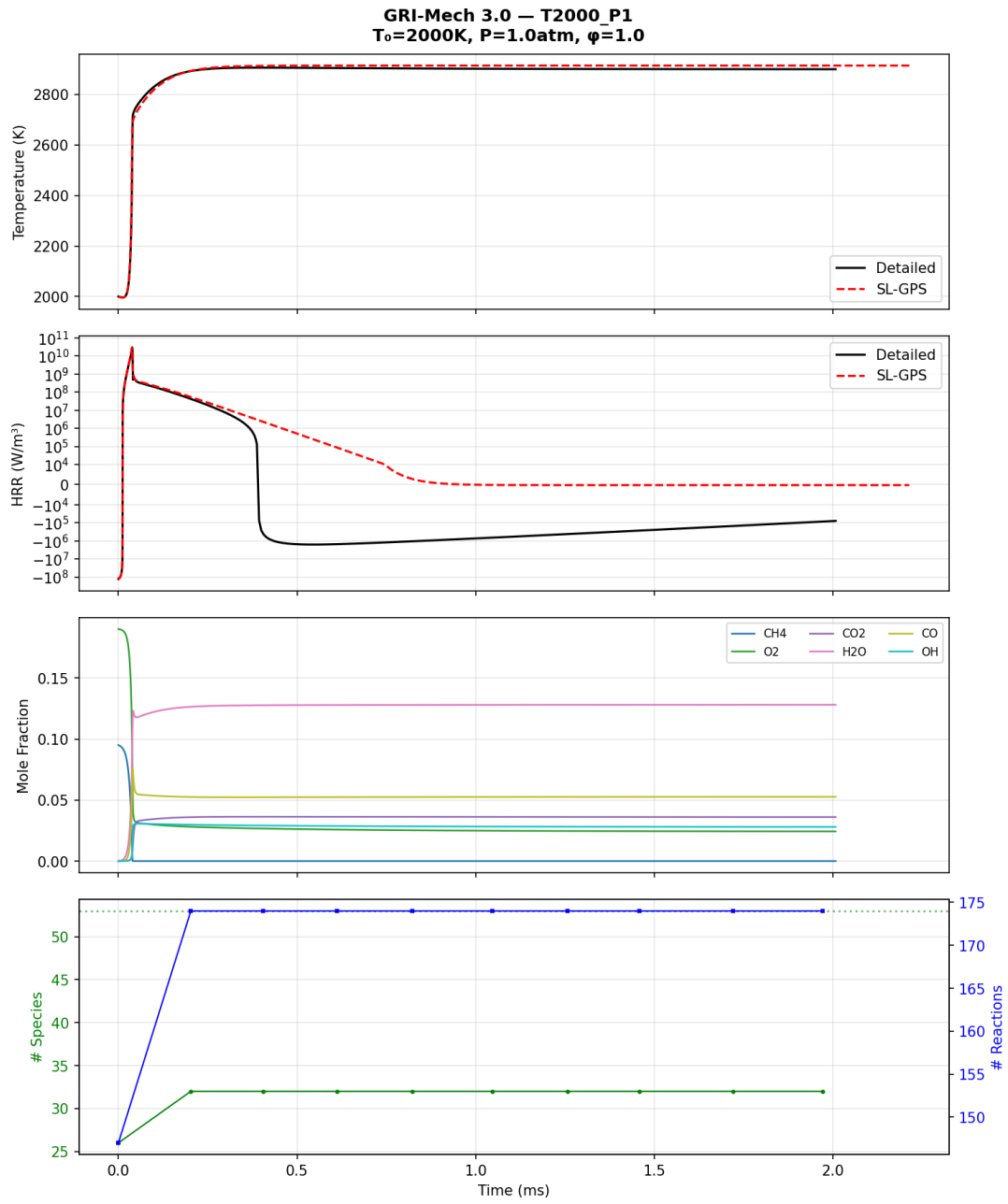


Figure 5: Case: T=2000K, P=1atm, ϕ =1.0

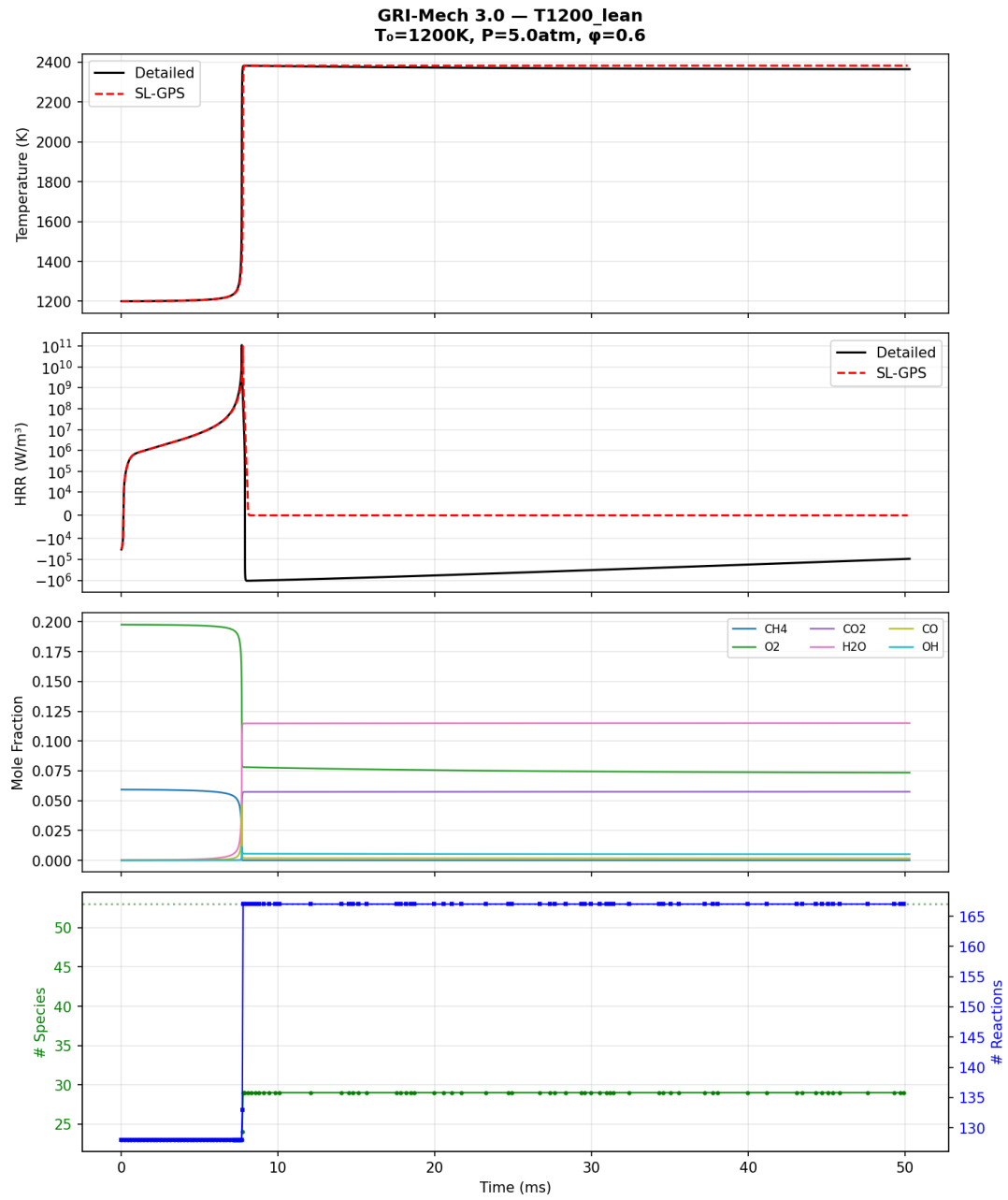


Figure 6: Case: $T=1200\text{K}$, $P=5\text{atm}$, $\phi=0.6$ (lean)